INJECTIVITY OF THE DIFFERENTIAL ON TANGENT SPACES IS EQUIVALENT TO UNRAMIFIEDNESS AT A POINT WITH EQUAL RESIDUE FIELDS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. This card was re-admitted after a gate audit: a text-similarity screen had flagged it against a statement of another chapter, but the two are different propositions, since that one characterizes unramifiedness of a morphism of algebraic spaces through the vanishing of the sheaf of differentials and formal unramifiedness, while this one is a tangent-space criterion at a point. The ruling recorded for this lane is therefore to keep this card.

Editorial note. The counted spans carry 4 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex), which since 2026-10-12 also carries the project's own reference/slogan/history definitions.

Editorial note. The label lemma-injective-tangent-spaces-unramified is defined only in varieties.tex, so no same-name restatement exists in another staged chapter.

Allowed standard prerequisites (editorial).

- Duality between the tangent space and the fibre of the cotangent module. Same chapter, Lemma lemma-map-tangent-spaces, invoked at proof lines 3162-3164. Supplies the translation from injectivity of the differential to surjectivity on the fibres of the differentials.
- Transitivity exact sequence for the module of differentials. Morphisms, Lemma morphisms-lemma-triangle-differentials, invoked at proof lines 3165-3166. Supplies the identification with the vanishing of the fibre of the relative differentials of f.
- Pointwise criterion for unramifiedness by vanishing of the fibre of the relative differentials. Morphisms, Lemma morphisms-lemma-unramified-at-point, invoked at proof lines 3169-3170. Supplies the last equivalence.
- Locally finite type is preserved by the morphism. Morphisms, Lemma morphisms-lemma-permanence-finite-type, invoked at proof lines 3160-3161. Supplies that the transitivity sequence applies.

Source reference. Original lemma 0B2F: sources/stacks-varieties/pinned-original-source.tex, source lines 3108–3108. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 01T8: sources/stacks-chow/tags.txt, source lines 2365–2365. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01UX: sources/stacks-chow/tags.txt, source lines 2426–2426. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02GF: sources/stacks-chow/tags.txt, source lines 3226–3226. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes locally of finite type over a base scheme S. Let $x \in X$ be a point. Set $y = f(x)$ and assume that $\kappa(y) = \kappa(x)$. Then the following are equivalent*

(1) *$df : T_{X/S,x} \longrightarrow T_{Y/S,y}$ is injective, and*
(2) *f is unramified at x.*

Proof. The morphism f is locally of finite type by Morphisms, Lemma 01T8. The map $\mathrm{d}f$ is injective, if and only if $\Omega_{Y/S,y} \otimes \kappa(y) \to \Omega_{X/S,x} \otimes \kappa(x)$ is surjective (Lemma 0B2F). The exact sequence $f^*\Omega_{Y/S} \to \Omega_{X/S} \to \Omega_{X/Y} \to 0$ (Morphisms, Lemma 01UX) then shows that this happens if and only if $\Omega_{X/Y,x} \otimes \kappa(x) = 0$. Hence the result follows from Morphisms, Lemma 02GF. □